PYTHON STUDY GROUP | PROJECT PROPOSAL

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Industry: Healthcare
Project Title: Improving Maternal Health Insights Through Data Analysis
Proposed Approach: Python-Based Analysis of Maternal Health Data

1. Problem Statement

Poor management of maternal health data can make it difficult for healthcare providers and health authorities to identify risks, monitor complications, assess outcomes, and plan services effectively. Incomplete, inaccurate, delayed, or disconnected records may obscure patterns in maternal morbidity and mortality, weaken continuity of care, and limit learning from adverse outcomes. Data problems do not cause every complication directly, but they can reduce the health system's ability to prevent, detect and respond to avoidable harm. This project will explore how Python can help turn maternal health data into clearer, more useful information for decision-making.

2. Project Aim and Objectives

Aim: To demonstrate how Python-based data cleaning, analysis and visualization can support better monitoring of maternal health outcomes.

- Inspect a suitable, de-identified or publicly available maternal-health dataset for missing values, duplicate records, inconsistent formats, and possible data-entry errors.
- Clean and prepare the data using Python libraries such as Pandas and NumPy, documenting assumptions and limitations.
- Analyze available indicators, such as antenatal visits, delivery outcomes, reported complications and maternal outcomes, to identify trends or patterns in the dataset.
- Create clear charts and a concise summary that could help stakeholders recognize data-quality gaps and areas needing further investigation.

3. Proposed Methodology

The team will first identify a suitable dataset from a reputable public source or use a synthetic dataset if real data cannot be accessed safely. We will review its structure and definitions, clean it, and calculate relevant summaries using Pandas and NumPy. Matplotlib may be used to visualize trends and data-quality issues. Findings will be interpreted cautiously, with attention to missing information, possible reporting bias and the fact that associations in the data do not prove causation. No identifiable patient information will be used.

4. How Python Can Contribute

Python can automate repetitive data-cleaning tasks, flag incomplete or inconsistent entries, summarize maternal-health indicators, compare patterns across time or facilities where the data permit, and generate charts or a simple dashboard. The project will not diagnose patients or predict individual outcomes. Instead, it will produce an exploratory analysis and practical recommendations for improving data quality and supporting evidence-informed planning.

5. Expected Output and Impact

Expected deliverables include a cleaned dataset, a reproducible Python notebook, visualizations, a short finding report and recommendations for improving maternal-health data recording and use. The project aims to demonstrate how better data quality and accessible analysis can help reveal service gaps, strengthen monitoring, and inform future investigation into preventable maternal morbidity and mortality.

Scope note: This is a data-analysis prototype, not a clinical tool. Its conclusions will depend on the quality, coverage, and definitions of the dataset available.